

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016–17

Zoology**Paper No. ZOB-501 : Functional Anatomy of Non-chordate**

Time: 3 hours

Full Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper. The figures in the right-hand margin indicate marks. Answer six questions including Question 1 which is compulsory.

Question 1. Answer the following parts:

(a) Define the following terms:

[4 × 1 = 4]

- (i) Stigma
- (ii) Scolex
- (iii) Medusa
- (iv) Gregarines.

(b) State whether the following statements are True or False. Give brief justification to your answer:

[4 × 2 = 8]

- (i) Gametocyst in *Monocystis* is generated from sporozoite.
- (ii) Fertilium is produced by male heteronereis.
- (iii) The correct sequence of larval stages in the development of *Fasciola* is Miracidia, Radiae, Sporocyst, Metacercaria and Cercaria.
- (iv) Surface tegument of *Fasciola* consists of microtriches for absorption of readymade food in host.

(c) Differentiate between the following:

[4 × 2 = 8]

- (i) Young and Mature trophozoite of *Monocystis*
- (ii) Bipinnaria and Pluteus
- (iii) Hydrozoa and Scyphozoa
- (iv) Aplacophoran and Polyplacophoran foot.

Question 2. Write the name of different types of canal system in sponges with example. Explain the Syconoid type of canal system with a neat diagram. $[2 + 8 = 10]$

Question 3. Describe the structure of *Euglena* with a neat diagram. Comment on euglenoid movement. $[7 + 3 = 10]$

Question 4. Describe the structure of polyp in *Obelia* with a neat diagram. $[10]$

Question 5. Discuss the structure and significance of Trochophore larva in Annelida. $[10]$

Question 6. Discuss the different types of larva in Echinoderms with labelled sketch. $[10]$

Question 7. Illustrate the life cycle of *Ascaris*. Mention the pathogenesis caused by *Ascaris*. $[8 + 2 = 10]$

Question 8. Write short notes on any two of the following in not more than 400 words each: $[2 \times 5 = 10]$

- (a) Theories of origin of Metazoa
- (b) Parasitic adaptation of *Fasciola*
- (c) Hydrostatic system of *Asterias*.

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